Functional Dependencies and Minimal Basis

1 Purpose

This document records the functional dependency identification and minimal-basis work for the Sprint Two goal *Calculate Minimal Bases and Project Relational Dependencies*. The projection and dependency-preservation portions of the goal will be documented separately. This analysis is based on the current Sprint Two ERD.

2 Updated Relational Schema Used for the Analysis

The updated ERD maps to the following relations:

$$\begin{aligned}
\text{Region}(&\textit{region_id}, \textit{region_name}, \textit{country})\\
\text{GPUType}(&\textit{gpu_type_name})\\
\text{SpecificGPUType}(&\textit{gpu_type_name}, \textit{memory_gb})\\
\text{AIModel}(&\textit{model_id}, \textit{model_name})\\
\text{GPUServer}(&\textit{machine_id}, \textit{cpu_capacity}, \textit{memory_capacity}, \textit{gpu_capacity},\\
&\textit{region_id}, \textit{gpu_type_name})\\
\text{InferenceRequest}(&\textit{request_id}, \textit{submitted_at}, \textit{input_size}, \textit{status},\\
&\textit{response_time_ms}, \textit{region_id}, \textit{model_id}, \textit{machine_id})\\
\text{MachineMetric}(&\textit{machine_id}, \textit{metric_id}, \textit{start_time}, \textit{end_time},\\
&\textit{worker_load}, \textit{gpu_utilization}, \textit{cpu_utilization})\\
\text{Hosts}(&\textit{machine_id}, \textit{model_id})\\
\text{CompatibleWith}(&\textit{model_id}, \textit{gpu_type_name})
\end{aligned}$$

The primary key of `MachineMetric` is now ($\textit{machine_id}, \textit{metric_id}$) because `MachineMetric` is a weak entity owned by `GPUServer`. The key of `SpecificGPUType` is `gpu_type_name`, it is also a foreign key to the parent `GPUType` relation.

Region role notation

The ERD has two different relationships to `Region`: a request originates in a region, while a server is located in a region. Both local SQL columns are named `region_id`, but they represent different roles. To avoid accidentally identifying them as the same attribute during the FD calculations, this document uses the temporary names

$$\textit{request_region_id} \qquad \text{and} \qquad \textit{server_region_id}.$$

When projected back onto the individual relations, both are written as the local name `region_id`.

3 Functional Dependencies from the ERD

The identifier of a strong entity functionally determines its non-identifier attributes. A many-to-one relationship also means the identifier on the many side determines the identifier on the one side. For the weak `MachineMetric` entity, the full identifier is the union of the owner identifier and the weak entity's local identifier. Before splitting right-hand sides, the non-trivial project dependencies are:

$$
\begin{aligned}
region_id &\to \{region_name, country\} \\
model_id &\to model_name \\
gpu_type_name &\to memory_gb \quad \text{(within SpecificGPUType)} \\
machine_id &\to \{cpu_capacity, memory_capacity, gpu_capacity, \\
&\qquad server_region_id, gpu_type_name\} \\
request_id &\to \{submitted_at, input_size, status, response_time_ms, \\
&\qquad request_region_id, model_id, machine_id\} \\
\{machine_id, metric_id\} &\to \{start_time, end_time, worker_load, \\
&\qquad gpu_utilization, cpu_utilization\}.
\end{aligned}
$$

`Hosts` and `CompatibleWith` are many-to-many relations with no non-key relationship attributes. Therefore neither relation contributes a non-trivial FD between its two participating identifiers. Their composite key constraints still remain part of the relational schema.

4 Minimal Basis

1. Split every FD so that each right-hand side contains one attribute.
2. Try removing each FD, if the remaining FDs still imply it, remove it.
3. Try removing attributes from the left-hand side of each FD; if the remaining left-hand side still implies the right-hand side, we will remove the extraneous attribute.

4.1 Step 1: Split the right-hand sides

After using the splitting rule, the candidate basis is:

$$
\begin{aligned}
region_id &\rightarrow region_name \\
region_id &\rightarrow country \\
model_id &\rightarrow model_name \\
gpu_type_name &\rightarrow memory_gb \\
machine_id &\rightarrow cpu_capacity \\
machine_id &\rightarrow memory_capacity \\
machine_id &\rightarrow gpu_capacity \\
machine_id &\rightarrow server_region_id \\
machine_id &\rightarrow gpu_type_name \\
request_id &\rightarrow submitted_at \\
request_id &\rightarrow input_size \\
request_id &\rightarrow status \\
request_id &\rightarrow response_time_ms \\
request_id &\rightarrow request_region_id \\
request_id &\rightarrow model_id \\
request_id &\rightarrow machine_id \\
\{machine_id, metric_id\} &\rightarrow start_time \\
\{machine_id, metric_id\} &\rightarrow end_time \\
\{machine_id, metric_id\} &\rightarrow worker_load \\
\{machine_id, metric_id\} &\rightarrow gpu_utilization \\
\{machine_id, metric_id\} &\rightarrow cpu_utilization.
\end{aligned}
$$

4.2 Step 2: Remove redundant FDs

Each FD was tested by temporarily removing it and calculating the closure of its left-hand side under the remaining dependencies. None of the removed right-hand-side attributes could still be derived, so no redundant FD was found.

For example, if

$$machine_id \rightarrow gpu_type_name$$

is removed, $\{machine_id\}^+$ under the remaining dependencies does not contain `gpu_type_name`. Therefore that FD must remain.

4.3 Step 3: Remove extraneous left-hand-side attributes

All determinants are single attributes except the weak entity dependencies for `MachineMetric`. For those dependencies:

- $\{metric_id\}^+$ does not determine any metric measurement attributes, so `machine_id` cannot be removed.
- $\{machine_id\}^+$ determines server information, but it does not determine a particular metric record's measurement attributes, so `metric_id` cannot be removed.

Therefore $\{machine_id, metric_id\}$ is minimal on the left-hand side.

4.4 Conclusion

After applying all three minimal-basis steps from lecture, the candidate basis from Section 4.1 satisfies the required conditions:

- every functional dependency has a single attribute on the right-hand side;
- removing any one of the listed dependencies prevents its right-hand-side attribute from being derived from the same determinant; and
- no attribute can be removed from the composite determinant $\{machine_id, metric_id\}$ without losing the corresponding `MachineMetric` dependency.

Thus no dependency is redundant and no determinant contains an extraneous attribute. The split dependency set listed in Section 4.1 is the project's minimal basis, denoted F_{min}. Projection of F_{min} onto the final relations will be handled in a separate document.